Virtual Machine Setup and Management

IT11 – Operating System Application

You can manage individual virtual machines or a group of virtual machines that belongs to a host or cluster. From the virtual machine's console, you can change the guest operating system settings, use applications, browse the file system, monitor system performance, and so on. Use snapshots to capture the state of the virtual machine at the time you take the snapshot.

To migrate virtual machines using cold or hot migration, including vMotion, vMotion in environments without shared storage, and Storage vMotion,

Installing a Guest Operating System

A virtual machine is not complete until you install the guest operating system and VMware Tools. Installing a guest operating system in your virtual machine is essentially the same as installing it in a physical computer.

Types of Installation

Using PXE with Virtual Machines

You can start a virtual machine from a network device and remotely install a guest operating system using a Preboot Execution Environment (PXE). You do not need the operating system installation media. When you turn on the virtual machine, the virtual machine detects the PXE server.

Types of Installation

Using PXE with Virtual Machines

PXE booting is supported for Guest Operating Systems that are listed in the VMware Guest Operating System Compatibility list and whose operating system vendor supports PXE booting of the operating system.

Types of Installation

Using PXE with Virtual Machines

The virtual machine must meet the following requirements:

Have a virtual disk without operating system software and with enough free disk space to store the intended system software.

Have a network adapter connected to the network where the PXE server resides

Types of Installation

Install a Guest Operating System from Media

You can install a guest operating system from a CD-ROM or from an ISO image. Installing from an ISO image is typically faster and more convenient than a CD-ROM installation. .

Types of Installation

Install a Guest Operating System from Media

Prerequisites:

- Verify that the installation ISO image is present on a VMFS datastore or network file system (NFS) volume accessible to the ESXi host.
- Alternatively, verify that an ISO image is present in a content library.
- Verify that you have the installation instructions that the operating system vendor provides.

Types of Installation

Upload ISO Image Installation Media for a Guest Operating System

You can upload an ISO image file to a datastore from your local computer. You can do this when a virtual machine, host, or cluster does not have access to a datastore or to a shared datastore that has the guest operating system installation media that you require.

Customizing Guest Operating Systems

IT11 – Operating System Application

When you clone a virtual machine or deploy a virtual machine from a template, you can customize the guest operating system of the virtual machine. You can change the computer name, network settings, and license settings.

Customizing guest operating systems helps prevent conflicts that occur if virtual machines with identical settings are deployed, for example conflicts due to duplicate computer names. You can apply customization as part of virtual machine deployment or later.

Microsoft Hyper V

IT11 – Operating System Application

Microsoft Hyper-V is Microsoft’s virtualization platform, or ‘hypervisor’, which enables administrators to make better use of their hardware by virtualizing multiple operating systems to run off the same physical server simultaneously.

Hyper V - Terminology

Hyper V

is the general product name for Microsoft’s hypervisor. It is available in a number of different offerings. When this word is used by itself, it applies to the hypervisor generically.

Hyper V - Terminology

Host

"Host" refers to a physical computer system that is running Hyper-V.

Hyper V - Terminology

Virtual Machine

A virtual machine, or “VM”, is a logical construct that is owned and operated by Hyper-V which presents the VM with “virtualized hardware.”

Hyper V - Terminology

Virtual Switch

Hyper-V does not allow its virtual machines to access the network directly. Instead, it creates Hyper-V virtual switches. This virtual switch is the intermediary between the physical network and the virtual network adapters used by virtual machines.

Hyper V - Terminology

Checkpoints

The complete running state of a virtual machine can be saved without interrupting its current operations. If necessary, those changes can be reverted to very quickly. Otherwise, they can be discarded with no impact.

Hyper V - Terminology

Save State

A virtual machine’s running state can be saved and then the virtual machine’s operations suspended. This allows its host to be powered off or for the virtual machine to be moved. The guest operating system does not know that anything has happened; it is simply notified of a large time change once it is turned back on.

Hyper V Manager

IT11 – Operating System Application

Hyper-V Manager is the most common tool used to interact with Hyper-V. It has these primary features

- Configure Hyper-V feature settings, such as Live Migration
- Create, destroy, and configure Hyper-V virtual machines
- Create, destroy, and manage virtual hard disks

Hyper-V Manager is the most common tool used to interact with Hyper-V. It has these primary features

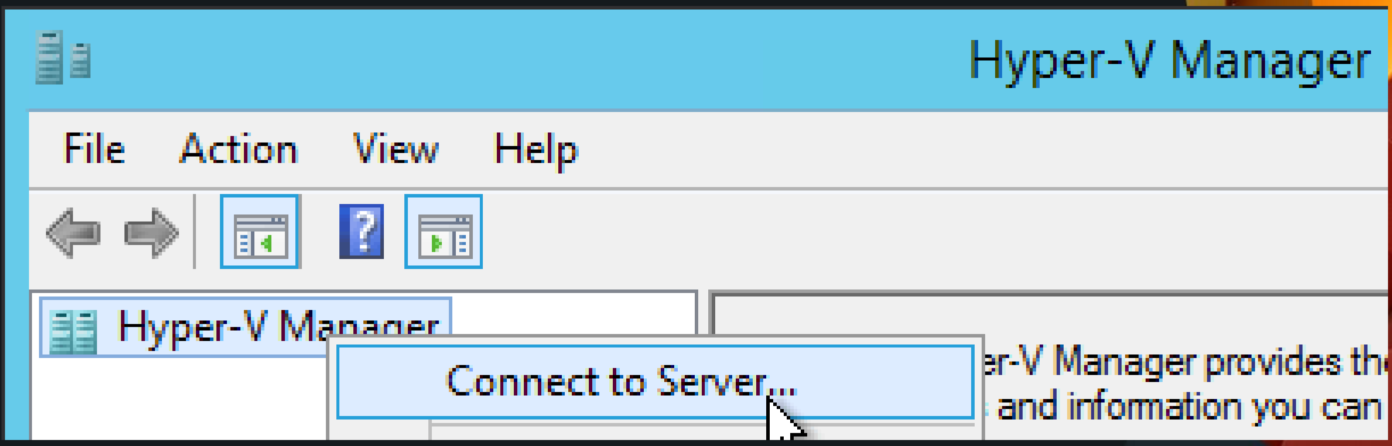
- Direct control of virtual machine functions; for example: stop, start, and save
- Create, delete, apply, and revert checkpoints
- Configure and control Hyper-V Replication

How to Connect to a Hyper-V Host

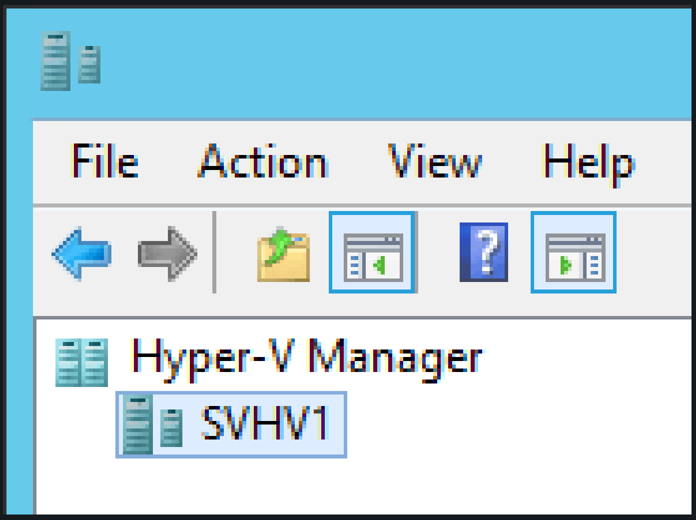
IT11 – Operating System Application

Upon opening Hyper-V Manager for the first time, the first thing to do is connect it to the system – if you’re operating it remotely. If Hyper-V is running locally, it will automatically include the local host.

Right-click on the Hyper-V Manager item in the left pane and click Connect to Server...

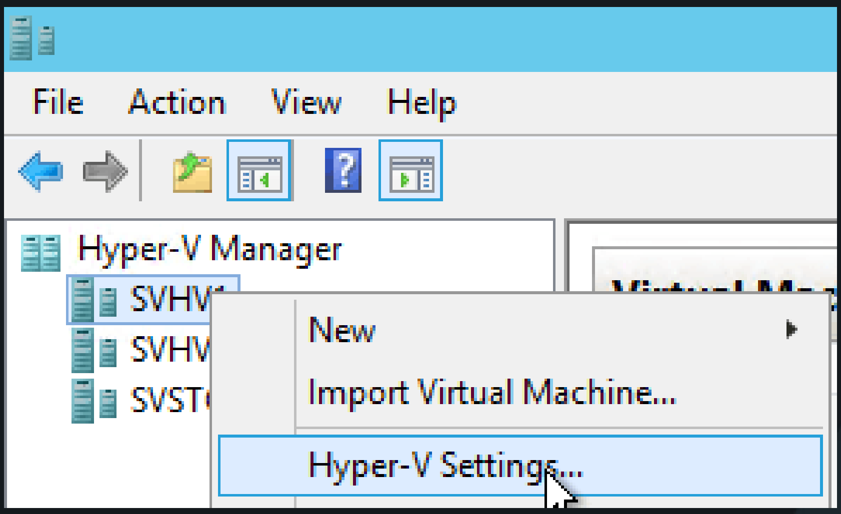


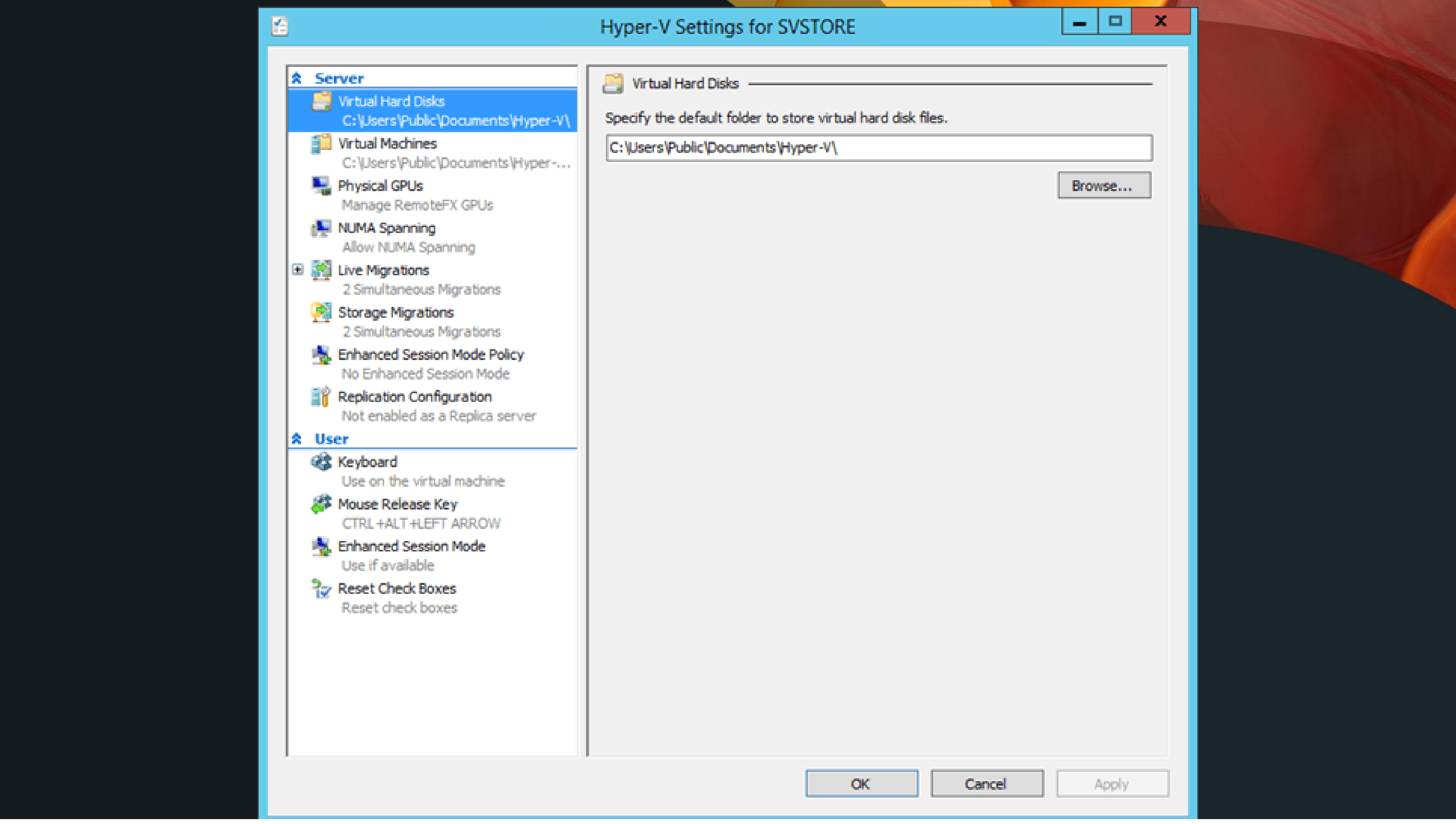
In the Select Computer dialog, enter the name or IP address of the remote Hyper-V system and click OK.



Configuring Basic Hypervisor Settings

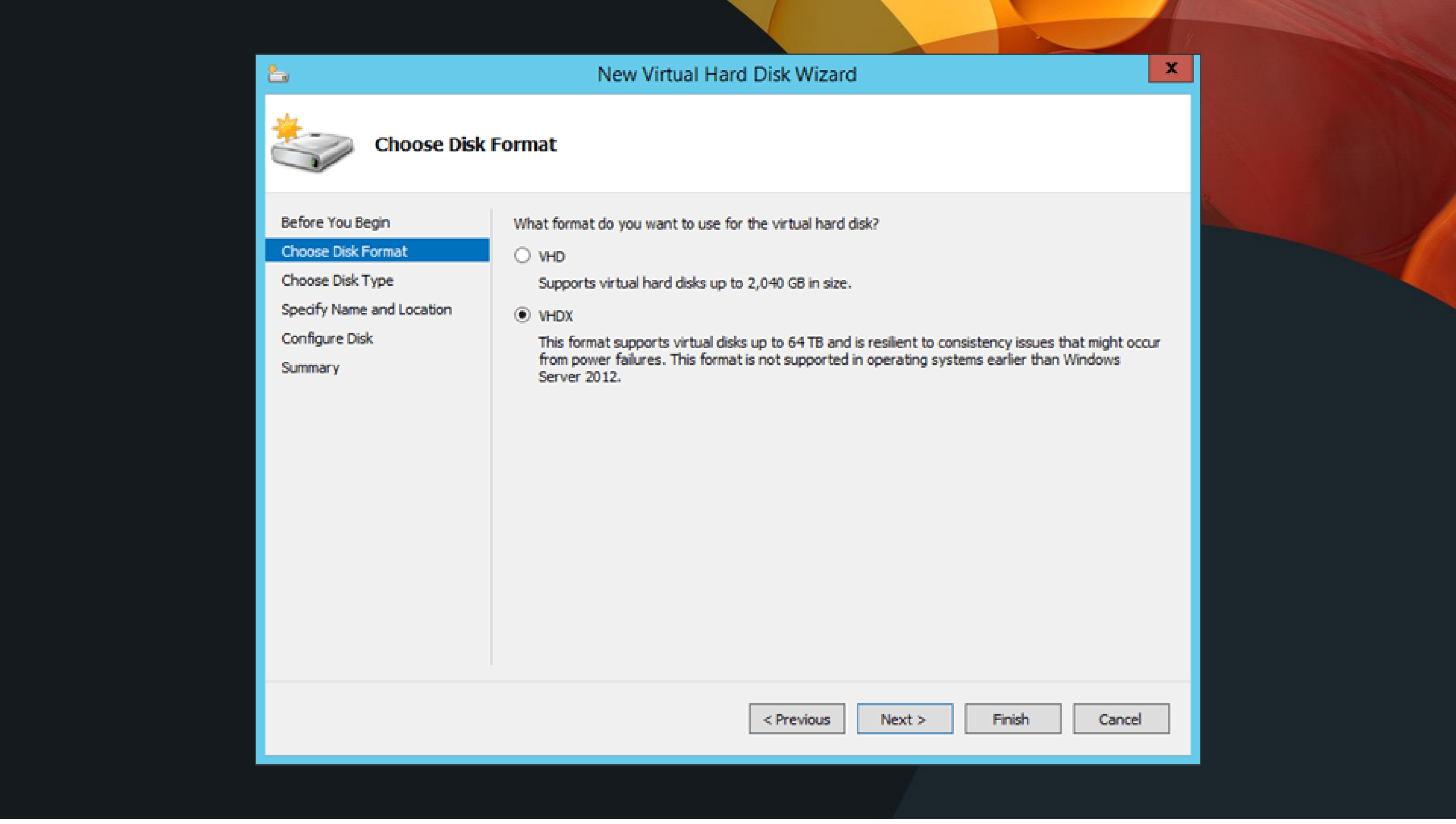
Right-click on the server in the left pane and click Hyper-V Settings:



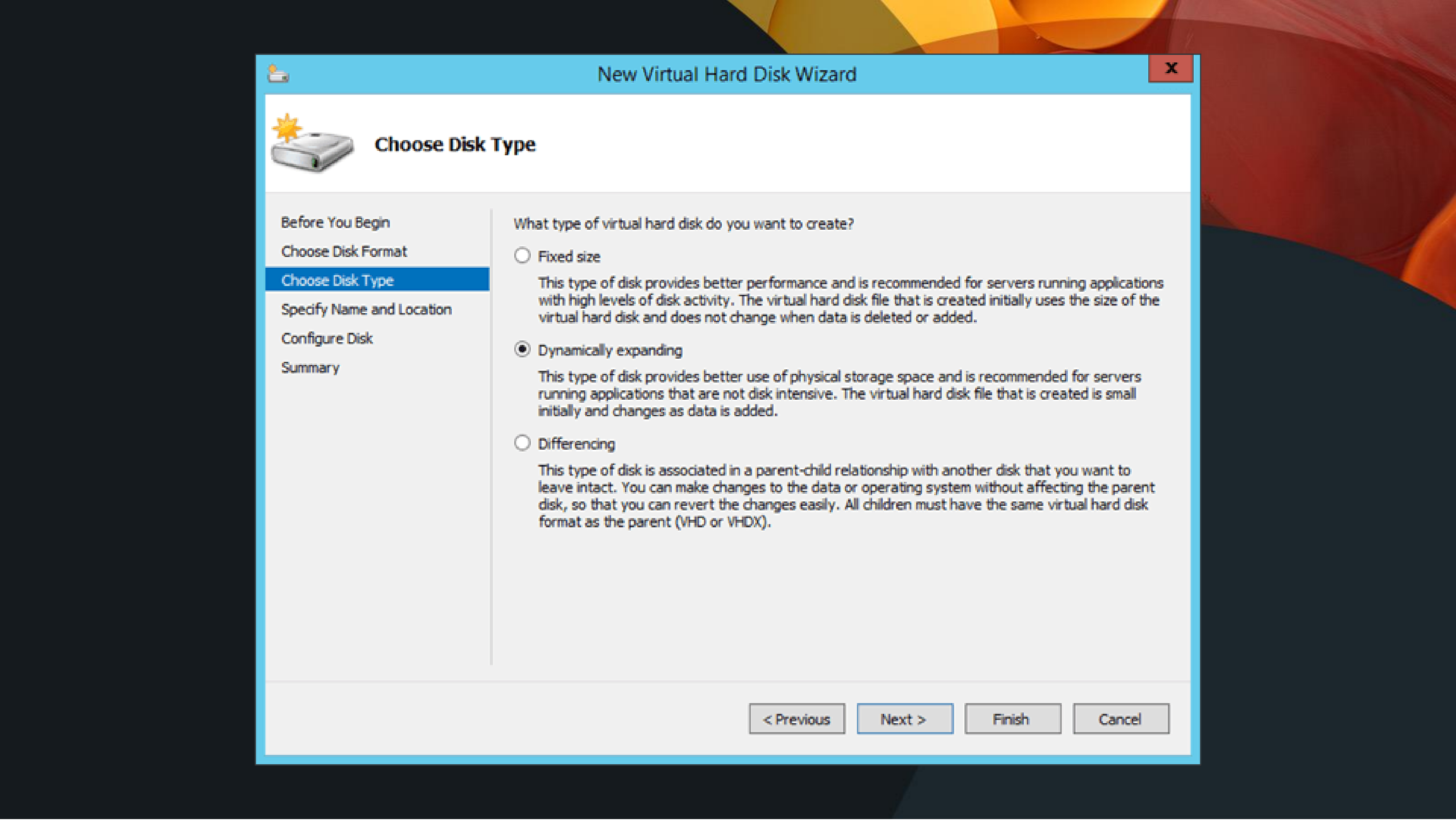

Hyper-V Settings for SVSTORE
Server
Virtual Hard Disks
C:\Users\Public\Documents\Hyper-V\
Virtual Machines
C:\Users\Public\Documents\Hyper-...
Physical GPUs
Manage RemoteFX GPUs
NUMA Spanning
Allow NUMA Spanning
Live Migrations
2 Simultaneous Migrations
Storage Migrations
2 Simultaneous Migrations
Enhanced Session Mode Policy
No Enhanced Session Mode
Replication Configuration
Not enabled as a Replica server
User
Keyboard
Use on the virtual machine
Mouse Release Key
CTRL+ALT+LEFT ARROW
Enhanced Session Mode
Use if available
Reset Check Boxes
Reset check boxes
Virtual Hard Disks
Specify the default folder to store virtual hard disk files.
C:\Users\Public\Documents\Hyper-V\
Browse...
OK
Cancel
Apply

How to Create Empty Virtual Hard Disks

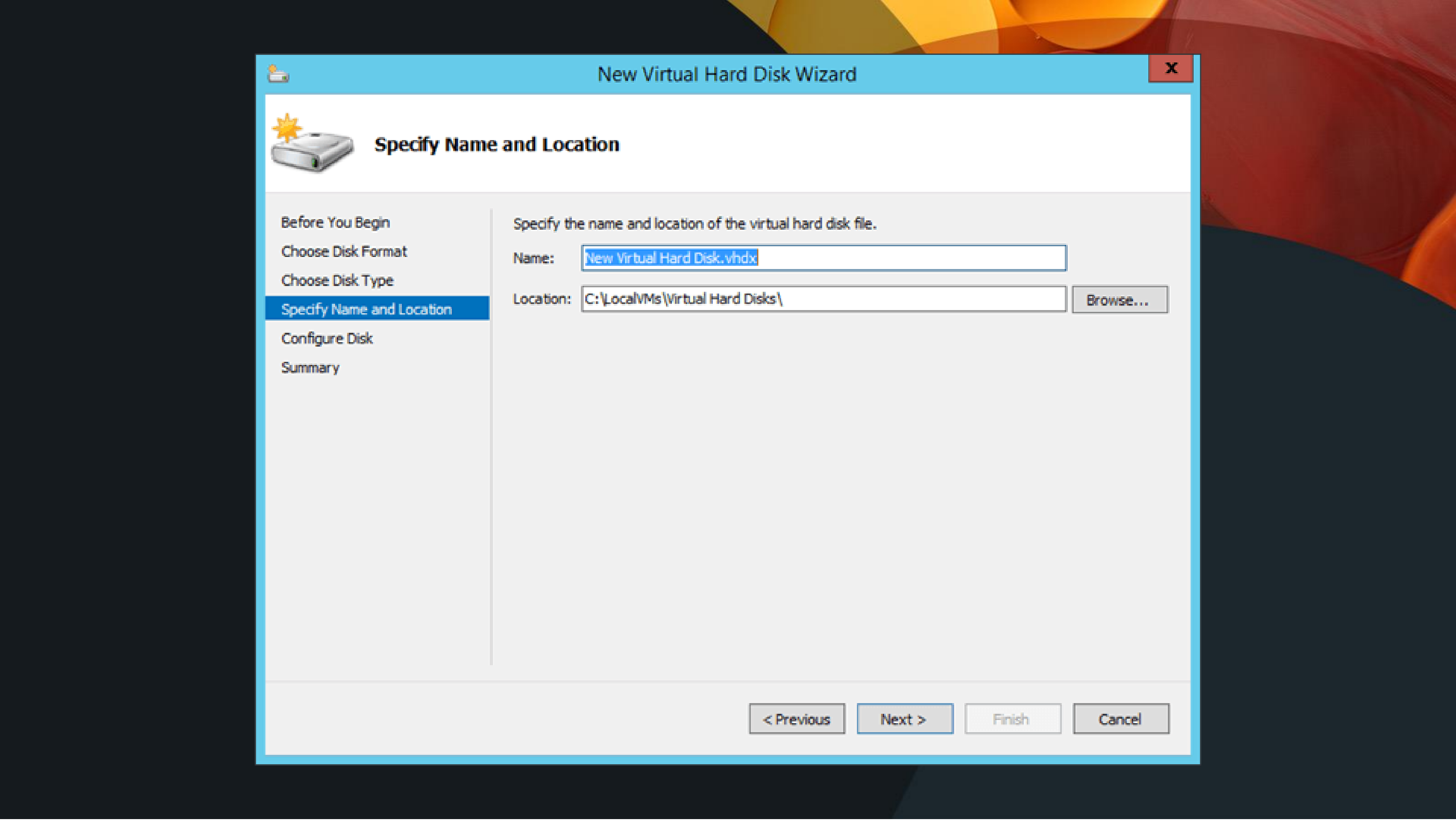
- On the host's action menu, select New and then Hard Disk… This opens the New Virtual Hard Disk wizard.
- The first screen is informational. Click Next.
- Choose whether to create a disk of the original VHD or the new VHDX type. Click Next.


New Virtual Hard Disk Wizard
Choose Disk Format
Before You Begin
Choose Disk Format
Choose Disk Type
Specify Name and Location
Configure Disk
Summary
What format do you want to use for the virtual hard disk?
VHD
Supports virtual hard disks up to 2,040 GB in size.
VHDX
This format supports virtual disks up to 64 TB and is resilient to consistency issues that might occur from power failures. This format is not supported in operating systems earlier than Windows Server 2012.
< Previous
Next >
Finish
Cancel

- Choose whether the disk should be Fixed, Dynamically Expanding, or Differencing. Dynamic is the default type and should be used unless you have a clear requirement for one of the others. Click Next.


New Virtual Hard Disk Wizard
Choose Disk Type
Before You Begin
Choose Disk Format
Choose Disk Type
Specify Name and Location
Configure Disk
Summary
What type of virtual hard disk do you want to create?
Fixed size
This type of disk provides better performance and is recommended for servers running applications with high levels of disk activity. The virtual hard disk file that is created initially uses the size of the virtual hard disk and does not change when data is deleted or added.
Dynamically expanding
This type of disk provides better use of physical storage space and is recommended for servers running applications that are not disk intensive. The virtual hard disk file that is created is small initially and changes as data is added.
Differencing
This type of disk is associated in a parent-child relationship with another disk that you want to leave intact. You can make changes to the data or operating system without affecting the parent disk, so that you can revert the changes easily. All children must have the same virtual hard disk format as the parent (VHD or VHDX).
< Previous
Next >
Finish
Cancel

- Choose the name and the location for the new virtual hard disk. Click Next.


New Virtual Hard Disk Wizard
Specify Name and Location
Before You Begin
Choose Disk Format
Choose Disk Type
Specify Name and Location
Configure Disk
Summary
Specify the name and location of the virtual hard disk file.
Name:
New Virtual Hard Disk.vhdx
Location:
C:\LocalVMs\Virtual Hard Disks\
Browse...
< Previous
Next >
Finish
Cancel

Fixed or Dynamically Expanding disks will ask you for the parameters of the disk to be created. You have three options.

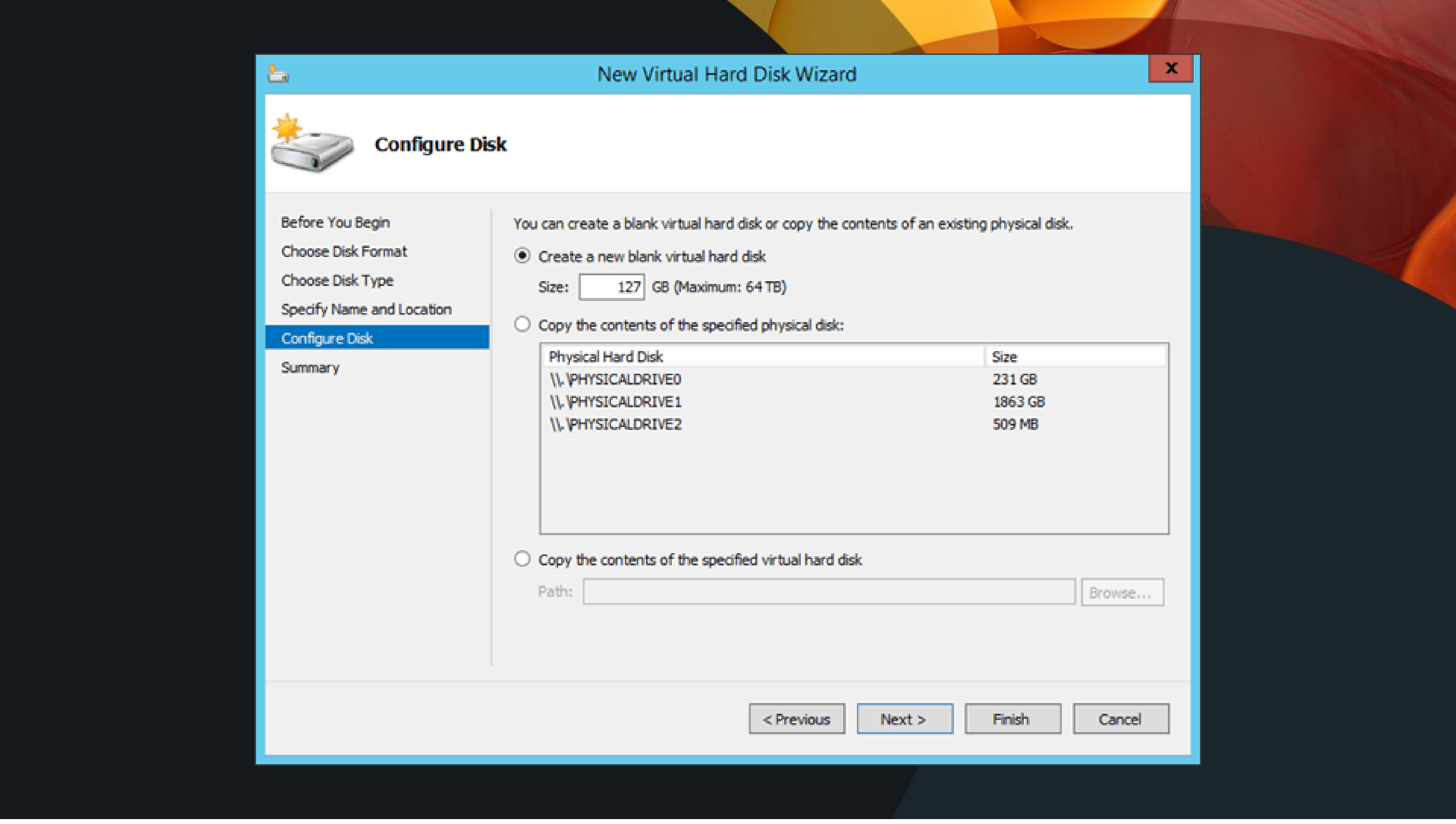
- Create a new blank virtual hard disk simply asks you to specify the size of the disk to create. If you are creating a Dynamically Expanding disk, this sets the upper limit on the file but creates it with only the basic header. For a Fixed disk, the entire file will be allocated and zero-filled.

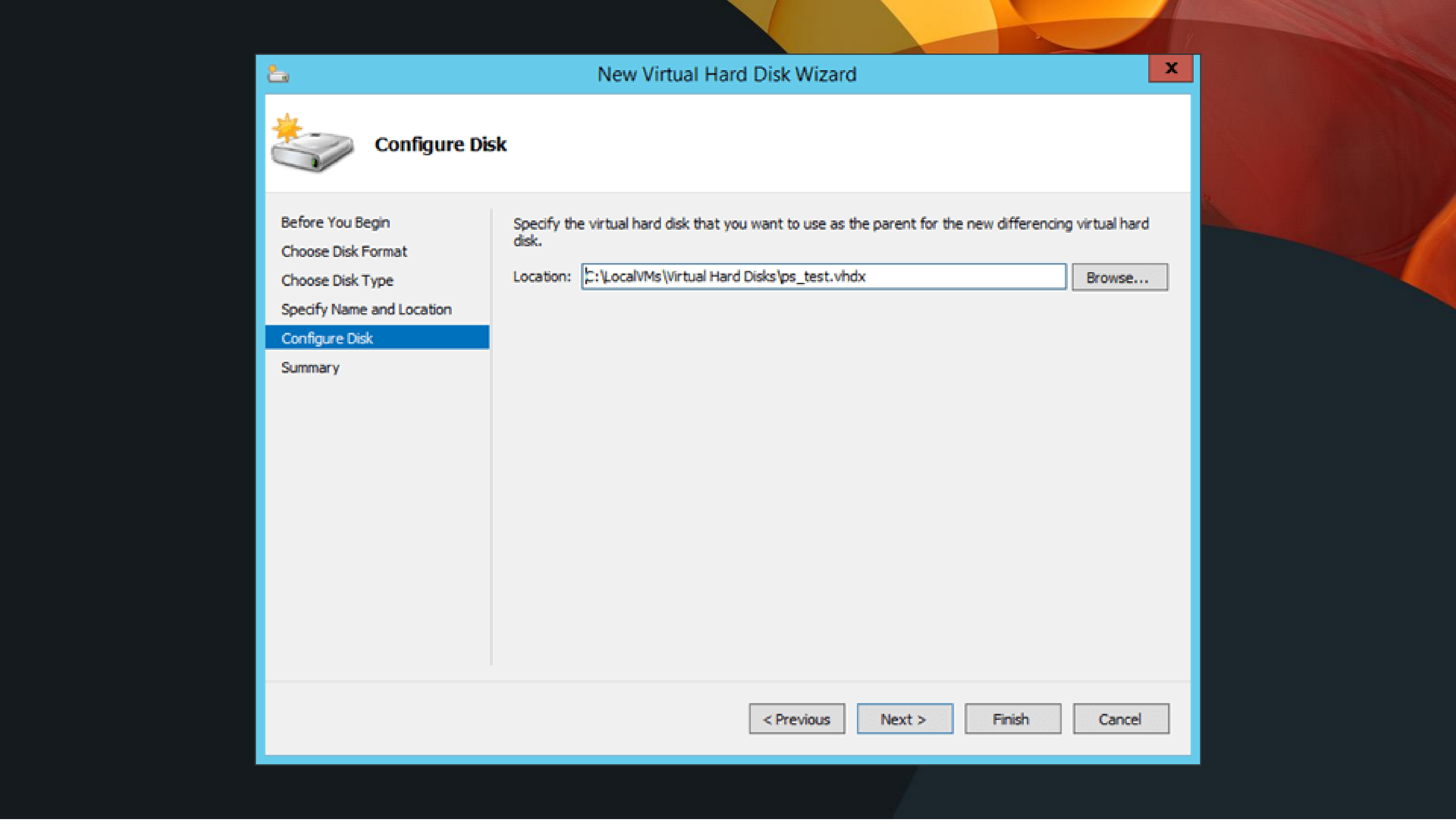
Fixed or Dynamically Expanding disks will ask you for the parameters of the disk to be created. You have three options.

- Copy the contents of the specified disk will duplicate the contents of the selected disk into a new file of the same size. If you selected it as a Dynamically Expanding disk, it will only allocate enough space to duplicate the allocated blocks of the source file.

Fixed or Dynamically Expanding disks will ask you for the parameters of the disk to be created. You have three options.

- Copy the contents of the specified hard disk will duplicate the contents of another virtual hard disk. This is a copy of contents, not a simple file copy of the original. So, if you selected to create a Dynamically Expanding disk, it may be smaller than the source.


New Virtual Hard Disk Wizard
Configure Disk
Before You Begin
Choose Disk Format
Choose Disk Type
Specify Name and Location
Configure Disk
Summary
You can create a blank virtual hard disk or copy the contents of an existing physical disk.
Create a new blank virtual hard disk
Size: 127 GB (Maximum: 64 TB)
Copy the contents of the specified physical disk:
Physical Hard Disk
Size
\\.\PHYSICALDRIVE0
231 GB
\\.\PHYSICALDRIVE1
1863 GB
\\.\PHYSICALDRIVE2
509 MB
Copy the contents of the specified virtual hard disk
Path:
Browse...
< Previous
Next >
Finish
Cancel


New Virtual Hard Disk Wizard
Configure Disk
Before You Begin
Choose Disk Format
Choose Disk Type
Specify Name and Location
Configure Disk
Summary
Specify the virtual hard disk that you want to use as the parent for the new differencing virtual hard disk.
Location:
C:\LocalVMs\Virtual Hard Disks\ps_test.vhdx
Browse...
< Previous
Next >
Finish
Cancel

References

https://docs.vmware.com/en/VMware-vSphere/7.0/com.vmware.vsphere.vm_admin.doc/GUID-B7023DD7-F790-4DF8-89B4-FF09DA3DBFB1.html
https://www.altaro.com/hyper-v/what-is-hyper-v/
https://www.altaro.com/hyper-v/hyper-v-manager-2/configuration/
https://www.altaro.com/hyper-v/hyper-v-manager-2/

Thank you!